# PROJECT SYNOPSIS

## PulseLink – Emergency Blood Donor Finder System

### 1. Introduction

Blood availability is an important factor during medical emergencies, accidents, surgeries, and other critical situations. Finding a suitable donor quickly can be difficult when information about available donors is scattered or outdated. PulseLink is a web-based mini project designed to help connect voluntary blood donors with people who need blood. It provides a central platform where users can register, search for donors by blood group and city, create blood requests, and respond to requests.

### 2. Abstract

PulseLink is an Emergency Blood Donor Finder System developed using Python Flask and SQLite. The system provides user registration and login, a personalized dashboard, donor search, blood-request management, and profile management. Users can search for potential donors using blood group and city filters, while people requiring blood can publish requests with details such as patient name, required blood group, units, urgency, city, hospital, and contact information. Registered users can volunteer to respond to requests and view responses related to their own requests. The project also includes password hashing and CSRF protection to improve application security. PulseLink is an educational demonstration project and is not a verified medical emergency service.

### 3. Problem Statement

During an urgent need for blood, patients and their families may have difficulty locating suitable donors within a short time. Manual searching through personal contacts or different online groups can be time-consuming. Therefore, a simple, organized, and accessible web application is needed to maintain donor information and blood requests in one place, allowing users to search and coordinate more efficiently.

### 4. Aim

To develop a web-based system that helps connect voluntary blood donors and people seeking blood by providing donor search, emergency blood-request posting, and response-management features.

# PROJECT DETAILS

## Objectives, Scope and Methodology

### 5. Objectives

- To provide a simple and responsive website for blood-donation coordination.
- To allow donors and recipients to register and securely log in.
- To enable donor search using blood group and city.
- To provide blood-group compatibility filtering as a search aid.
- To allow users to create and view active blood requests.
- To classify requests as Normal, Urgent, or Critical.
- To let volunteers respond to requests and let request creators view responses.
- To allow users to manage their profiles and donor availability.
- To store application data in an SQLite database and protect important forms and passwords.

### 6. Scope of the Project

The system includes a landing page, registration and login pages, a personalized dashboard, a donor directory, blood-request listings, a request-creation form, and a profile-management page. Users can search by blood group and city, check available donor profiles, publish requests, respond to requests, and update request status. The application is intended for learning and demonstration. It does not verify donor eligibility, guarantee donor availability, or replace professional medical advice or emergency services.

### 7. Proposed Methodology

1. Requirement Analysis: Identify the needs of donors and people requesting blood, including registration, searching, requests, and responses.

2. System Design: Plan the website pages, navigation, database tables, and user workflow.

3. Front-End Development: Create responsive web pages and forms for registration, login, dashboard, donor search, requests, and profiles.

4. Back-End Development: Use Python Flask to process form submissions, manage sessions, validate inputs, and handle application logic.

5. Database Integration: Use SQLite to store user profiles, donor details, blood requests, and responses.

6. Security Implementation: Apply password hashing, input validation, session handling, and CSRF protection.

7. Testing: Check registration, login, donor filtering, request creation, response submission, profile updates, and error handling.

8. Deployment/Demonstration: Run the application in a suitable development environment and demonstrate its main features.

### 8. System Workflow

A user registers or logs in and reaches the dashboard. A user seeking blood can create a request by entering the required details. Donors can search for requests or search the donor directory using blood group and city. Interested volunteers can respond to requests. The person who created a request can review responses and update the request status. Profile settings allow users to maintain their information and availability.

# IMPLEMENTATION & CONCLUSION

## Modules, Technology and Expected Outcome

### 9. Main Modules

- Landing Page: Introduces PulseLink and directs users to the main features.
- Registration Module: Collects account and basic donor/recipient information.
- Login and Logout Module: Provides access to registered users and ends sessions securely.
- Dashboard Module: Shows useful account information, quick actions, and recent blood requests.
- Find Donors Module: Filters donor listings by blood group and city.
- Blood Requests Module: Displays active requests and their urgency levels.
- Create Request Module: Records patient/request details, required blood group, units, city, hospital, contact number, and notes.
- Response Module: Allows volunteers to respond and request creators to view responses.
- Profile Module: Lets users update personal details, donor status, and availability.
- Database and Security Module: Uses SQLite, password hashing, validation, and CSRF protection.

### 10. Tools and Technologies

Programming Language: Python

Back-End Framework: Flask

Database: SQLite

Front-End: HTML, CSS, and responsive web design (with JavaScript if used in implementation)

Security: Password hashing, session handling, input validation, and CSRF protection

Platform: Web browser; suitable for demonstration in a local development environment

### 11. Expected Outcome

The completed system is expected to provide a clear and user-friendly way to manage donor information and blood requests. It should help users find listed donors using relevant filters, publish requests with urgency information, and coordinate responses through one application. The project also demonstrates the practical use of Python Flask, database operations, web forms, authentication, and basic web-security practices.

### 12. Conclusion

PulseLink is a web-based Emergency Blood Donor Finder System intended to make blood-donation coordination more organized and accessible. By combining donor search, request posting, volunteer responses, profile management, and database storage, the application demonstrates how a full-stack web project can address a meaningful real-world problem. In future versions, the system could be improved with verified donor records, location-based search, notifications, stronger privacy controls, and integration with authorized healthcare organizations. Any real-world use would require appropriate verification and safeguards.

### Project Details (to be completed)

Student Name: _____

Class / Semester: _____

College: _____

Project Guide: _____

Academic Year: _____